

Jacob Horne

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EDUCATION

University of California, Irvine **June 2028**
Bachelor of Science in Computer Science; Campuswide Honors Collegium **GPA: 3.92**
Coursework: Object-Oriented Programming (C/C++), Data Structures & Algorithms, Databases, Computer Architecture

EXPERIENCE

Sandia National Laboratories **Aug 2026 - Present**
Software Engineering Intern - AI R&D Tooling **Livermore, CA**
- Building AI R&D tooling that makes Sandia's AI research usable by researchers, engineers, and mission users.
- Turning research workflows into practical software interfaces, tooling, and systems users can rely on.

Capital One **Jun 2026 - Aug 2026**
Software Engineering Intern - ML Recommender Infrastructure **New York, NY**
- Replaced a network API call with an in-process SDK to reduce latency by 25% for 60M+ customers monthly.
- Built Databricks/PySpark sampling pipeline over 800M+ historical customer-message outcomes, enabling candidate recommender models to test against broader production behavior 28x faster.
- Automated sampling and upstream jobs in Apache Airflow with 18 schema, distribution, and multiplier quality checks.

UCI Digital Learning Lab **Dec 2025 - Present**
Research Assistant - LLM Systems and Evaluation **Irvine, CA**
- Built Python-based multi-agent essay-grading pipelines with LangGraph, implementing critic, advocate, and judge agents.
- Analyzed LLM output datasets using NumPy, Pandas, and Matplotlib for token log-probs, behavior, and confidence.
- Co-authored paper accepted to ICLR 2026 LLM Workshop on multi-agent evaluation and token-level LLM confidence.

Calit2 **Apr 2026 - Present**
AI/ML Undergraduate Researcher - Physics-Informed Machine Learning **Irvine, CA**
- Built and optimized a physics-informed PyTorch/GRU time-series pipeline with heat-transfer residual losses and initial-condition constraints to replace hours-long FEA simulations with fast cylindrical wall temperature predictions.
- Accelerated model training with MPS/GPU support, gradient clipping, bounded parameter discovery, and multi-seed benchmarking to compare PINN, LSTM, and GRU architectures across timestep resolutions and thermal scenarios.
- Achieved R-squared = 0.99 with 12,000x faster inference than FEA, enabling scalable heat-transfer modeling.

CareTech, UCI **Oct 2025 - Jun 2026**
Software Developer - Computer Vision **Irvine, CA**
- Built a Next.js/FastAPI food-scanning app for real-time model inference and automated vitamin and nutrient tracking.
- Integrated PyTorch inference with backend APIs and nutrition data, converting food predictions into live intake totals.
- Processed and augmented 28K+ food images across 270+ categories, training PyTorch models to 81% accuracy.

Commit the Change, UCI **Oct 2025 - Jun 2026**
Full-Stack Developer - Backend and Platform **Irvine, CA**
- Built Node.js/Express APIs processing 5,000+ weekly requests for a scheduling platform serving 18,000+ patients.
- Reduced booking conflicts and manual scheduling workload by 30% with real-time appointment and quota tracking.

PROJECTS

Agonus (Blockchain, UCI) | LangChain, FastAPI, Next.js, Solidity, PostgreSQL **Oct 2025 - Apr 2026**
- Built a decentralized AI trading platform where autonomous agents compete in crypto tournaments with on-chain betting, real-time performance tracking, tokenized reward incentives, and a live tournament dashboard UI.
- Architected LangChain ReAct agents with custom tools for market data, trade execution, planning, and memory; designed Celery/Redis orchestration with scheduling, locking, and crash recovery to manage 10+ concurrent agents.

Tata Consultancy Services Capstone | FastAPI, PostgreSQL, pgvector, OpenAI, DeBERTa **Jan 2026 - Jun 2026**
- Co-developed a React/FastAPI banking chatbot with DeBERTa ethics checks for safe, compliant customer guidance.
- Owned an end-to-end document RAG pipeline with PDF/OCR ingestion, document classification, figure extraction, OpenAI embeddings, PostgreSQL/pgvector retrieval, and banking-metric analysis to ground responses in customer data.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C/C++, SQL, Java, R, MIPS Assembly
ML/Data: PyTorch, scikit-learn, TensorFlow, PySpark, OpenCV, LangChain, LangGraph, NumPy, Pandas, Matplotlib
Platforms/Tools: Databricks, Airflow, PostgreSQL, Redis, Celery, FastAPI, Docker, Git, AWS (ECS, Fargate, S3, CloudWatch)